

Task 3.1P

Solve the following set of problems using Python and submit the code file with extension .ipynb.

1. Download and read the "[Real estate valuation data set.xlsx](#)" dataset. You can find the description of the features and targets on the UCI repository website.
2. Split the dataset in train and test set (use your choice of splitting). Train a linear regression model and report the performance (use your choice of at least three performance metrics).
3. Continue from question 2, use leave-one-out validation and five fold cross validation and report the model performance. Compare the three evaluation methods and suggest which one is more reliable.
4. Continue from question 2, apply L1 and L2 regularization on the linear regression model developed using the same training set. Compare the performance and indicate key findings.

Assessment feedback

The results with comments will be released within 5 business days from the due date.

Referencing

You must correctly use the IEEE method in this assessment. See the [referencing guide](#).

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